

Moment RAG vs Chunk RAG on McRaven's "Make Your Bed"

Same transcript, same embedder, same answer model, same 12 questions. The only variable is the retrieval unit: fixed chunks or semantic moments.

Kurt Lozier · September 25, 2026 · Option 1: Brief Presentation

4.58 vs 4.58

answer quality, moment vs baseline
LLM judge (Haiku), not human

0.71 vs 0.52

context purity: share of retrieved
words inside the right lesson

921 vs 1202

mean context tokens per question
(23% fewer)

\$0.0043 vs \$0.0054

answer cost per question
(20% lower)

After three versions, Moment RAG ties the baseline on answers and wins on context size and purity.

V1 · DRIFT ONLY

Embedding drift sets boundaries

Haiku may only merge. Stories split from their morals.

boundary score **0.60** · purity **0.32** · judge **4.42**

V2 · LLM-BOUNDED

Haiku moves boundaries ± 3 sentences

Split and merge allowed. Story and moral rejoin.

boundary score **0.90** · purity **0.37** · judge **4.67**

V3 · ADAPTIVE K

Keep extra moments only when they score near the top

1.92 moments per question instead of 3.

boundary score **0.90** · purity **0.71** · judge **4.58**

V3 RESULT

	BASELINE	MOMENT	READ
Answer quality · LLM judge (Haiku), not human, 1–5	4.58	4.58	tie
Citation accuracy	0.93	0.92	tie
Context purity · context tokens · answer cost	0.52 · 1202 · \$0.0054	0.71 · 921 · \$0.0043	moment
Lesson recall	0.94	0.83	baseline

A talk with **ten numbered lessons** gives moment boundaries an objective answer key.

19.4 min

Adm. William H. McRaven, UT Austin
commencement, 2014 (YouTube yaQZFhrW0fU)

281

manual English caption segments, about 3,200
words

Why this video

- The refrain marks where each lesson ends.
- **ground_truth.py** finds all ten refrains; starts hand-checked.
- A lesson runs from its story's first sentence to its refrain.
- Boundaries get a score, not an eyeball.

196

sentences after splitting: the unit both pipelines
share

10

lessons, each closed by "if you want to change
the world ..."

Ground truth · data/lessons.json

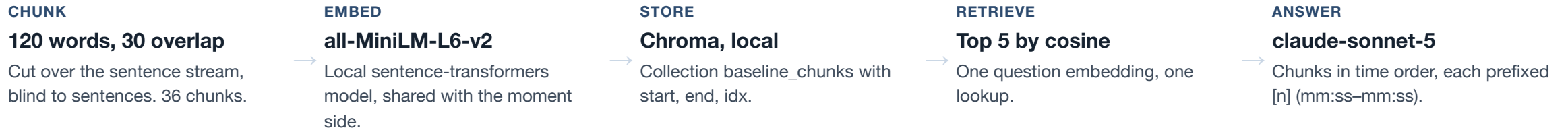
#	LESSON	START	#	LESSON	START
1	Make your bed	4:44	6	Slide down head first	11:30
2	Find help to paddle	6:13	7	Don't back down from sharks	12:52
3	Size of their heart	7:20	8	Best in the darkest moment	14:02
4	Sugar cookie	8:38	9	Sing in the mud	15:38
5	Circuses	10:05	10	Never ring the bell	17:50

The reference **builds moments at query time**; this build persists them once at ingest.

	REFERENCE CODEBASE (MOMENTSEARCH)	THIS BUILD
What a moment is	Not stored. A transient group of hits.	A stored span with title and summary.
How it forms	Frame hit + transcript hit within a 15 s fusion window	Embedding drift, then one Haiku refine-and-label pass
LLM at ingest	None. Fetch, sample frames, CLIP and bge embeddings.	One Haiku call for the whole video
Retrieval	RRF (k=60) over frame and text branches; ×1.5 cross-modal boost; top 6	RRF (k=60) over summary and full-text collections; adaptive $k \leq 3$
LLM at query	One cited answer; abstains below a confidence gate	One cited answer (Sonnet), same as the baseline
Eval pattern	Fixed golden set, recall@10, human citation sheet	Fixed golden set, swap only the retrieval layer

Write path vs read path. The reference spends its moment work on every query. This build moves all of it to ingestion, so the query path costs the same as chunk RAG.

The baseline cuts **fixed 120-word chunks** and ignores sentence and story boundaries.



Answer rules, shared by both pipelines

- Cite every claim with **[n]** and a timestamp.
- No outside knowledge; the excerpts only.
- Every call logs tokens, seconds and dollars.

The designed weakness

- Lessons run about 300 words, chunks 120.
- A lesson's story and its moral land in different chunks.
- Top 5 buys coverage with off-topic words.

Embedding drift proposes boundaries; one Haiku pass turns them into 14 labelled moments.

STAGE 1 · NO LLM

Drift candidates

3-sentence windows; TextTiling depth > mean + 0.5·std; min 6 sentences. 17 segments; 6 of 10 lesson starts within 10 s.

STAGE 2 · ONE HAIKU CALL

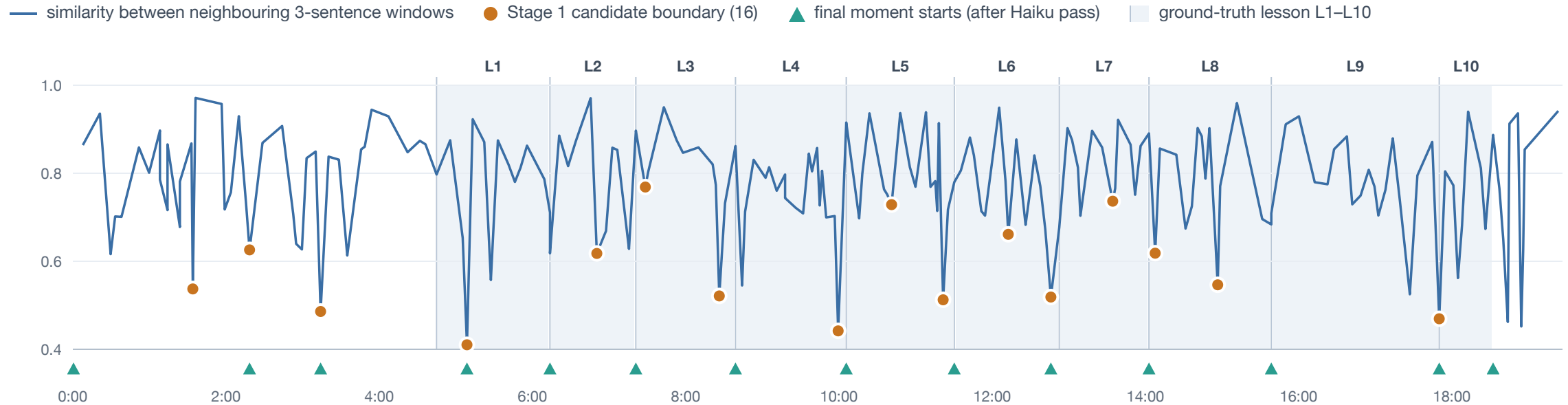
Refine and label

Move $\leq \pm 3$ sentences, split, merge sparingly. Code validates; one retry. 14 moments; 9 of 10 starts (0.90).

STAGE 3 · INDEX

Two collections

moment_summaries (title + summary) and moment_text (full text).



Dips mark topic shifts. Candidates often land 1–2 sentences early, pushing a moral into the next segment; Stage 2 repairs most of them: 9 of 10 final moment starts sit within 10 s of a lesson start (L1 is the miss). Source: data/drift.json, data/moments.json, data/lessons.json.

RRF over two collections plus an **adaptive cut** returns 1.92 moments per question.

How a query runs

- Embed the question once.
- Query summaries and full text, top 10 each.
- Fuse with Reciprocal Rank Fusion, $k=60$.
- Always keep the top moment.
- Keep moments 2–3 only if $\text{score} \geq 0.97 \times \text{top}$.
- Sonnet reads full moment text, labelled [n] Title (mm:ss–mm:ss).

Offline sweep, moment side (n=12)

CONFIG	RECALL	PURITY	MOMENTS
fixed k=3 (v2)	0.83	0.37	3.00
R = 0.5 to 0.9 (planned grid)	0.83	0.37	3.00
fixed k=2	0.83	0.61	2.00
R = 0.97 (chosen)	0.83	0.71	1.92
R = 0.98	0.83	0.75	1.75
R = 0.985	0.76	0.89	1.17
fixed k=1	0.64	0.84	1.00

Disclosure: R was tuned on the same 12 questions; there is no held-out set. RRF compresses scores (moments 2–3 score ≥ 0.95 of the top everywhere), so the planned grid changed nothing. 0.97 sits on a plateau; 0.98 is one step from the recall cliff at 0.985.

Twelve questions in **three types** test single facts, story-plus-moral, and cross-lesson answers.

LOOKUP · 4

Answer sits inside one lesson

"How long was the rope on the slide for life obstacle?"

SPAN · 4

Needs the story and its moral

"What happened to a student who failed the uniform inspection, and what does McRaven say that teaches?"

MULTI · 4

Spans two or three lessons

"What do the night swim and the underwater ship attack have in common, and how do their lessons differ?"

METRIC	DEFINITION
lesson_recall	Share of expected lessons covered by at least one retrieved unit
context_precision	Share of retrieved units that cover an expected lesson
context_purity	Word-weighted share of retrieved context inside an expected lesson
citation_accuracy	Share of distinct [n] citations whose unit covers an expected lesson
phrase_coverage	Share of 3 key phrases present verbatim in the answer
judge_score	LLM judge (Haiku), not human: answer vs reference answer, 1–5

Covers: a unit overlaps $\geq 50\%$ of the lesson, or $\geq 50\%$ of the unit lies inside it. Precision and purity both rise when fewer units return; purity is the fairer of the two.

Moments match answer quality with **purier, smaller context**, but miss more lessons.

METRIC (OVERALL MEAN, N=12)	BASELINE (K=5 CHUNKS)	MOMENT V3 (K ≤ 3)	BETTER
judge_score · LLM judge (Haiku), not human, 1–5	4.58	4.58	tie
citation_accuracy	0.93	0.92	tie
context_purity	0.52	0.71	moment
context_precision	0.57	0.68	moment
context_tokens (mean)	1202	921	moment
latency_s (mean)	4	3	moment
answer_usd / question	\$0.0054	\$0.0043	moment
lesson_recall	0.94	0.83	baseline
phrase_coverage	0.75	0.67	baseline

Noise floor. The baseline retrieves the same chunks every run, yet its judge moved 4.42–4.58 and phrase coverage 0.75–0.81. Moment deltas smaller than that are not signal.

Each version fixed one failure the previous eval exposed.

MOMENT SIDE, OVERALL (N=12)	V1	V2	V3	BASELINE
moments · k	11 · 3	14 · 3	14 · ≤ 3	36 chunks · 5
boundary score (lesson starts within 10 s of a moment start)	0.60	0.90	0.90	–
lesson_recall	0.83	0.83	0.83	0.94
context_purity	0.32	0.37	0.71	0.52
citation_accuracy	0.82	0.86	0.92	0.93
judge_score · LLM judge (Haiku), not human	4.42	4.67	4.58	4.58
context_tokens (mean)	1670	1447	921	1202
answer usd / question	\$0.0063	\$0.0056	\$0.0043	\$0.0054

v1 → v2: boundaries

- Ship-attack story and moral rejoin one moment.
- q10 judge rises from 2 to 5.

v2 → v3: context size

- Adaptive k drops weak extra moments.
- Purity 0.37 → 0.71; tokens 1447 → 921.

One moment answered the sugar-cookie question with **half the tokens** of five chunks.

q06: "What happened to a student who failed the uniform inspection, and what does McRaven say that teaches?"

Baseline · 5 chunks, stitched from [2] [3] [4]

purity **0.52** tokens **1237** cost **\$0.0062** judge **5**

"According to excerpt [2] (8:26–9:11) and [3] (8:53–9:43), a student who failed the uniform inspection had to 'run, fully clothed into the surf zone' ... an effect known as a 'sugar cookie.' ... As stated in excerpt [4] (9:26–10:16), the lesson is: '... If you want to change the world, get over being a sugar cookie and keep moving forward.'"

- Story and moral sit in three overlapping chunks.
- Two retrieved chunks are off-lesson (4:07, 16:00).

Moment v3 · 1 moment: "Lesson four" (8:39–10:06)

purity **1.00** tokens **601** cost **\$0.0038** judge **5**

"According to excerpt [1] (8:39–10:06), a student who failed the uniform inspection had to run 'fully clothed into the surf zone,' ... an effect known as a 'sugar cookie.' ... the lesson for succeeding, and for changing the world, is to 'get over being a sugar cookie and keep moving forward' [1]."

- Story and moral arrive as one unit.
- Adaptive k kept a single moment.

Most moment misses are **boundary or breadth** problems, not retrieval noise.

QUESTION	SCORE, MOMENT VS BASELINE	CAUSE
q01 lookup	recall 0.00 vs 1.00; judge 5 vs 5	Boundary miss: sentences 43–45 stayed in the intro moment. Right answer, scored as a miss.
q11 multi	recall 0.33 vs 0.67; judge 2 vs 3	Adaptive k cut to one moment. "Numbers" share no vocabulary across lessons.
q09 multi	recall 0.67 vs 0.67; judge 4 vs 3	Lesson 9 (mud flats) missed; the baseline also missed one lesson.
q12 multi	purity 1.00 vs 0.58; judge 4 vs 5	Perfect retrieval; the answer hedged. Wording, not retrieval.
Distractor	extra unit in q07, q08, q10	"Find help to paddle" summary echoes the talk's theme, so it matches many questions.

The clearest cost of adaptive k is q11: questions that enumerate across the talk need breadth that the score gap does not signal.

Every call is costed, and all moment work sits on the write path.

Cost lines

ITEM	USD
Ingestion: Haiku labelling, 14 moments, 2 calls	\$0.0253
Ingestion per moment	\$0.0018
Answer per question, baseline	\$0.0054
Answer per question, moment v3	\$0.0043
LLM judge (Haiku) per answer judged	\$0.0008
Full eval v1 / v2 / v3	\$0.1608 / \$0.1502 / \$0.1370
Total spend, every run	\$0.5270

Rates: Sonnet 5 \$2 / \$10 and Haiku 4.5 \$1 / \$5 per 1M input / output tokens.

Cohort 1 reviewer feedback, carried forward

Jayita Chatterjee, Cohort 1 reviewer, 2026-08-07

- **Cost instrumentation:** a price table; every call returns usd.
- **Write vs read path:** moments persist at ingest; queries stay cheap.
- **Query path** = baseline's: 1 embedding, 2 lookups, 1 Sonnet call.
- **Worker scaling:** a limitation here, not built.
- Ingest runs sequential and single-process today.

Source: README.md "Cost"; reports/scorecard.md; DESIGN.md "Carried forward from Cohort 1 A3 reviewer feedback". Total includes a smoke run estimated at about \$0.0171.

The tie is real but small; a held-out set and a human pass come next.

LIMITATION	NEXT STEP
No held-out set; R tuned on the 12 questions	Write 12+ held-out questions before viewing v3 retrievals
Judge is Haiku, not human	Human faithfulness pass on all 24 v3 answers
One 19-minute video; one labelling prompt	Windowed labelling with overlap; validate across seams
Weak adaptive-k signal: RRF squeezes scores to 0.95–1.0	Try raw cosine or smaller RRF k; floor of 2 for enumerations
Stage 2 skipped a fix (q01)	Offline check: flag moments near a strong drift dip
Validator gap: a 4-sentence move passes as merge + split	Track moves and splits separately
Worker scaling: sequential, single-process ingest	Parallel embedding and labelling workers; write path only

Finding: better units did not raise answer scores on a 19-minute talk, but they cut context by 23% and cost by 20% at zero extra query-time work.